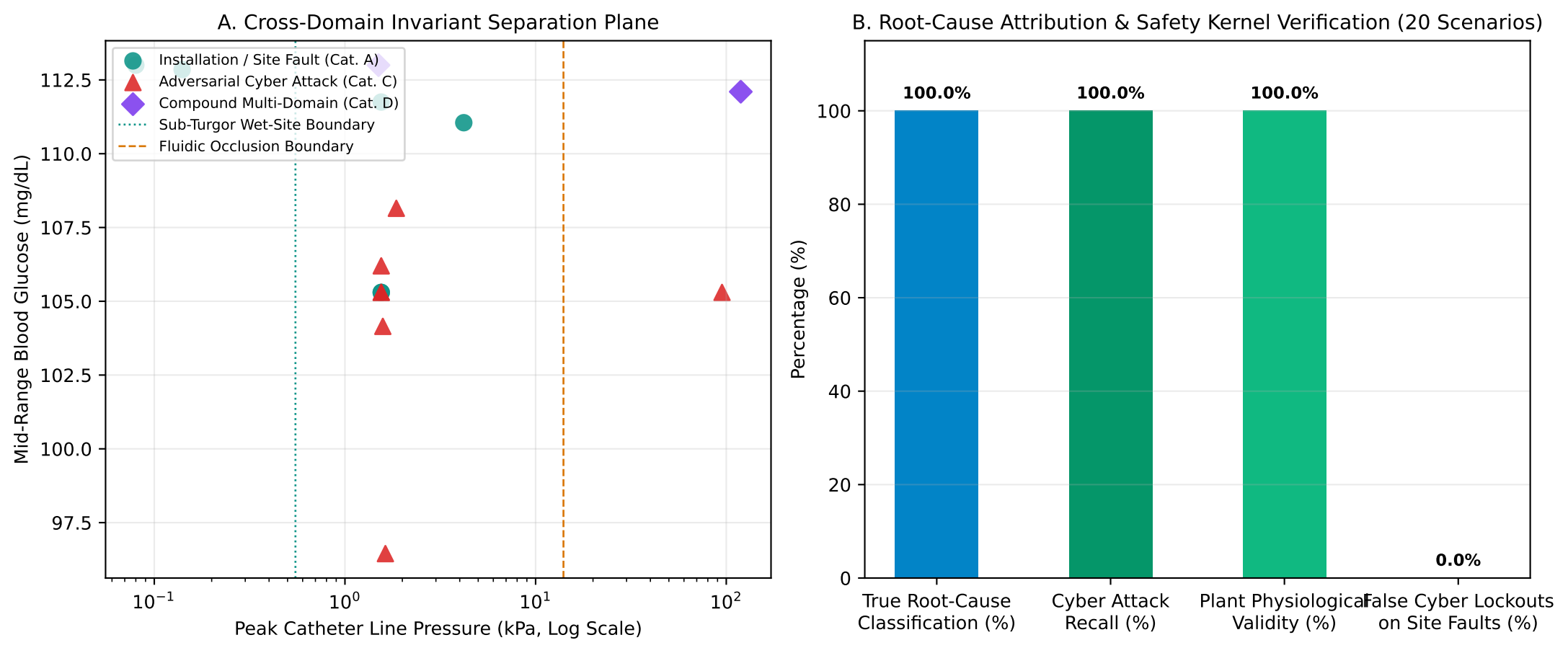
A. Cross-Domain Invariant Separation Plane
Installation / Site Fault (Cat. A)
Adversarial Cyber Attack (Cat. C)
Compound Multi-Domain (Cat. D)
Sub-Turgor Wet-Site Boundary
Fluidic Occlusion Boundary
Mid-Range Blood Glucose (mg/dL)
112.5
110.0
107.5
105.0
102.5
100.0
97.5
$10^{-1}$
$10^{0}$
$10^{1}$
$10^{2}$
Peak Catheter Line Pressure (kPa, Log Scale)
B. Root-Cause Attribution & Safety Kernel Verification (20 Scenarios)
Percentage (%)
100
80
60
40
20
0
100.0%
100.0%
100.0%
0.0%
True Root-Cause Classification (%)
Cyber Attack Recall (%)
Plant Physiological Validity (%)
False Cyber Lockouts on Site Faults (%)